# California Environmental Atlas

HackRPI 2026 • Codex handoff • November 7, 2026

## Product concept

An address-first interactive map that reveals the environmental, public-health, community, and essential-infrastructure context around any California location. The core experience is visual exploration—not a long home-risk report. Different selections automatically change the map, legend, and contextual data panel.

## User experience

Search a California address → fly to it in a polished 3D basemap → identify and outline its census tract → choose one of four modes → switch indicators to animate the relevant choropleth or infrastructure markers. Show the selected tract's raw value, California percentile when available, measurement definition, source year, and source link. Keep the address pinned as modes change.

## Four map modes

HEALTH | Asthma emergency visits, cardiovascular/heart-attack emergency visits, diabetes prevalence, and low-birth-weight rates. Display census-tract shading with a compact health summary.
ENVIRONMENT | PM2.5, ozone, diesel particulates, traffic impacts, toxic air releases, and cleanup-site burden. Tract choropleths are primary; optional verified site markers are a later enhancement.
COMMUNITY | Poverty, educational attainment, housing burden, and age/racial-ethnic population breakdowns. Use neutral charts and colors; demographics are not hazards.
INFRASTRUCTURE | Actual mapped hospitals/healthcare facilities, public schools, and power plants. Show clickable point markers and straight-line distances. Add fire services only if a reliable feed is found.

## Data: keep it small and credible

CalEnviroScreen 5.0 (CES 5.0): one California census-tract dataset supplies nearly everything in Health, Environment, and Community, including tract geometry and percentile fields. Prefer downloading and preprocessing it rather than calling the ArcGIS service on every map interaction.
California Department of Public Health (CDPH): healthcare facility locations. California Department of Education (CDE): public-school locations. California Energy Commission (CEC): power-plant locations. Cache or bundle filtered geographic extracts for reliable demos.

Verified starting links: CES 5.0 ArcGIS layer; CES 5.0 state download; CDPH healthcare GIS; CDE school sites GIS; CEC power plants GIS. Inspect live schemas, data dates, usage terms, and response formats before writing adapters.

# Implementation pipeline

1. Ingest CES 5.0 tract boundaries and indicators; retain stable tract IDs, raw values, percentiles, units, years, and nulls. Reproject to WGS84. Prepare simplified vector tiles/PMTiles for smooth statewide rendering.
2. Import healthcare, schools, and power plants as a separate point-layer dataset; retain names, categories, coordinates, status/date where available, and source URLs.
3. Geocode an entered address to latitude/longitude, run a point-in-polygon lookup for its tract, then retrieve its indicator record. Explicitly handle inaccurate geocodes and missing matches.
4. Build Next.js + TypeScript + Tailwind frontend with MapLibre GL JS for 3D basemap, fly-to animations, tract outlines, choropleths, facility markers, tooltips, and a persistent side panel. Turf.js can handle client-side spatial operations.
5. Define a single layer registry (mode, field, label, unit, value/percentile, source/year, visualization type, legend) so all four modes use the same rendering components rather than four separate map implementations.

# MVP and success criteria

Must work: a real California address resolves to the correct tract; switching Health, Environment, and Community visibly changes real CES map layers; Infrastructure shows at least healthcare and school pins (power plants next); any selected layer displays an intelligible legend, values, source, and missing-data state. Keep polished animated transitions central to the demo.
Stretch: compare two addresses using the same metrics with synchronized maps; add verified remediation-site pins and more infrastructure categories. Do not prioritize a nationwide rollout or dozens of unrelated datasets.

# Accuracy rules and differentiator

CES figures describe census tracts, not an individual property or person. Asthma and cardiovascular values are population-level indicators, not personal disease probabilities. Avoid causal claims, fabricated risk scores, or treating percentiles as rates. Show measurement definitions and timeframes.
Differentiation from Zillow and the state CES dashboard: cinematic, address-first navigation; four intelligently coordinated map modes; real critical-infrastructure overlays; and intuitive location comparison—not simply repackaging scores into a report.

# Codex: first engineering task

Verify one CES 5.0 ArcGIS point query against a known California address, fetch its tract attributes, and render a working selectable choropleth with one infrastructure point source. Establish the data contract and layer registry before investing in advanced animations or secondary integrations.